

Unit III: Linear Algebra

Assignment 1

- 1 If $f(t)$ is a fixed polynomial in t , of degree k and let P_n be the vector space of all polynomials in t of degree n . Show that $T : P_n \rightarrow P_{n+k}$ given by $T(p(t)) = p(t)f(t)$ is a linear transformation.
- 2 What are the rank, range, null space and nullity of T where T is as in question 1 above?
- 3 For what constants of a, b, c, d is the map $T : \mathbb{R}^3 \rightarrow \mathbb{R}$ given by $T(x, y, z) = ax + by + cz + d$ a linear transformation?
- 4 For all such T in Question 3 above which are linear transformations, for what values of a, b, c is the rank of T equal to 1?

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Assignment 2

- 1 Suppose S is rotation by an angle t , in $\mathbb{R}^2$. T and U are reflections in $\mathbb{R}^2$ in the lines $y = x$ and x -axis respectively. If $S = TU$ what is t ?
- 2 If T and U are the reflections above, is $TU = UT$? Justify your answer.
- 3 Let S be rotation by a right angle, anticlockwise, T be the reflection in the line $y = -x$ and U be the reflection in x -axis. Compare the linear transformations TS , ST and U .

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Assignment 3

- 1 Given the matrix for the projection onto the line L spanned by the vector v , what is the matrix for the reflection in L ? (HINT: the line joining any vector w with its reflection Rw has as midpoint the projection Pw , of w onto L .)
- 2 Project the vector $(1, -4, 3)$ onto the line spanned by the vector $(2, 5, 6)$. Why do we get such a result?
- 3 Project the vector $(7, 8, 9)$ onto the line spanned by $(1, 0, 3)$. What if $(1, 0, 3)$ were replaced by $(2, 0, 6)$? justify your answer.
- 4 Answer question 1 above where L is replaced by the column span of a matrix A .

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Assignment 4

- 1 Find the equation of the plane in the real 3-d space orthogonal to the intersection of the xz-plane with the plane $x - 2y + 3z = 0$.
- 2 Find a basis for the plane in the real 4-d space, orthogonal to the vectors $(1,2,3,4)$ and $(1,-2,3,-4)$.
- 3 What is the dimension of the space of all vectors in the real 5-d space orthogonal to the vectors $(4,5,2,3,1)$ and $(6,9,7,8,0)$?

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Assignment 5

- 1 Find the matrix M which projects every point in $\mathbb{R}^5$ onto the plane P given by $x+y+z-2w-3t=0$.
- 2 Find the line of best fit through the data points $(11,2)$, $(19,3)$, $(31,4)$ and $(39,5)$.
- 3 Split the vector $(1,2,3,4,5)$ as a sum of a vector from P and a vector from $P^\perp$ where P is the plane in question 1 above.